

Decarbonizing French Aviation: Addressing CO₂ Utilization & Modal Shift Challenges

Team Information:

Team Number: 264

Team Members: Anexy Johnney, Cilgin Cyriac, Sayanth Kakkarayil, Devaraj Ramamoorthy, Krutik Pravin-chandra Vanjara

Vision Behind This Action

Goal: Transform France's aviation sector into a **low-carbon, sustainable industry** by 2040, leveraging CO₂ utilization and shifting short-haul travel to rail.

Current Scenario (2024): The Challenges

- France captures **4.7 Mt CO₂** annually but **99% is used for enhanced oil recovery (EOR)**, locking in fossil dependency.
- Only **1%** of captured CO₂ is used for **Sustainable Aviation Fuel (SAF)**, despite the potential to meet **30% of aviation fuel demand**.
- **Flights emit 72x more CO₂** per person than trains, yet **68% of business travelers prioritize speed over sustainability**.

Implemented Strategy (2024–2040)

- **CO₂ Utilization Reform:** Mandate **30% of captured CO₂ for SAF** by 2030.
- **Nuclear-DAC Synergy:** Deploy **Direct Air Capture (DAC)** at nuclear plants for 24/7 CO₂ capture.
- **CO₂ Market Creation:** Establish **€200/t carbon pricing for SAF** on Euronext.
- **Rail & Modal Shift:** Expand TGV routes, impose kerosene taxes, and build airport-train hubs to increase rail share.

Future Scenario (2040): A Sustainable Aviation Industry

Expected Outcomes in 2040

- **100%** of domestic aviation CO₂ is recycled.
- **90% rail dominance** for short-haul travel, reducing domestic flights.
- **SAF production reaches 1.5 Mt/year**, covering 50% of aviation fuel needs.
- Aviation emissions reduced by **92% compared to 2025 levels**.

Key Milestones

Year	Action	Expected Impact
2025–2027	Carbon Valorization Act, launch 2 DAC pilot projects	10% CO ₂ diverted to SAF
2028–2035	Scale nuclear-DAC hubs, retrofit 20 industrial plants	50% CO ₂ used for SAF
2036–2040	Full SAF transition for domestic flights, rail expansion	100% aviation CO ₂ recycled

A Call to Action

The transition to sustainable aviation is possible, but requires commitment from policymakers, industry leaders, and citizens. By investing in CO₂ utilization, promoting high-speed rail, and taxing high-emission flights, France can set a global example in decarbonized aviation.